

3761. Minimum Absolute Distance Between Mirror Pairs

Solved 

Medium

 Topics

 Companies

 Hint

You are given an integer array `nums`.

A **mirror pair** is a pair of indices (i, j) such that:

- $0 \leq i < j < \text{nums.length}$, and
- `reverse(nums[i]) == nums[j]`, where `reverse(x)` denotes the integer formed by reversing the digits of `x`. Leading zeros are omitted after reversing, for example `reverse(120) = 21`.

Return the **minimum** absolute distance between the indices of any mirror pair. The absolute distance between indices `i` and `j` is `abs(i - j)`.

If no mirror pair exists, return `-1`.

Example 1:

Input: `nums = [12,21,45,33,54]`

Output: 1

Explanation:

The mirror pairs are:

- $(0, 1)$ since `reverse(nums[0]) = reverse(12) = 21 = nums[1]`, giving an absolute distance `abs(0 - 1) = 1`.
- $(2, 4)$ since `reverse(nums[2]) = reverse(45) = 54 = nums[4]`, giving an absolute distance `abs(2 - 4) = 2`.

The minimum absolute distance among all pairs is 1.

Example 2:

Input: `nums = [120,21]`

Output: 1

Explanation:

There is only one mirror pair $(0, 1)$ since `reverse(nums[0]) = reverse(120) = 21 = nums[1]`.

The minimum absolute distance is 1.

Example 3:

Input: `nums = [21,120]`

Output: `-1`

Explanation:

There are no mirror pairs in the array.

Constraints:

- `1 <= nums.length <= 105`
- `1 <= nums[i] <= 109`

Python:

class Solution:

def minMirrorPairDistance(self, nums: List[int]) -> int:

res = 100000

seen = {}

for i, n in enumerate(nums):

if n in seen:

res = min(res, i - seen[n])

seen[int(str(n)[::-1])] = i

return -(res == 100000) | res

JavaScript:

/**

* @param {number[]} nums

* @return {number}

*/

var minMirrorPairDistance = function(nums) {

let res = 100000;

let i = 0;

const seen = new Map();

```

    for (let n of nums) {
        if (seen.has(n))
            res = Math.min(res, i - seen.get(n));
        let r = 0;
        for (; n; n = (n / 10) | 0)
            r = r * 10 + (n % 10);

        seen.set(r, i++);
    }

    return -(res === 100000) | res;
};

```

Java:

```

class Solution {
    public int minMirrorPairDistance(int[] nums) {
        int res = 100000, i = 0;
        HashMap<Integer, Integer> seen = new HashMap<>();

        for (int n : nums) {
            int r;
            if (seen.containsKey(n))
                res = Math.min(res, i - seen.get(n));

            for (r = 0; n > 0; n /= 10)
                r = r * 10 + (n % 10);

            seen.put(r, i++);
        }

        return res == 100000 ? -1 : res;
    }
}

```